

Final Report

Homework 2

FINM 25000

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Strategy Descriptions With Entry and Exit Rules

Strategy 1 – Trend Following

The Trend Following strategy attempts to capture sustained upward price movements by combining momentum and trend-strength indicators. The ADX is used to confirm the trend is strong enough to trade, while the MACD is used to identify specific reversal or continuation triggers (like momentum shifts and pullbacks). Although not used directly in the trading logic, the price chart also displays the 20-day SMA as a visual reference for short-to-medium-term price trends.

The strategy uses:

- MACD (Moving Average Convergence Divergence) - measures momentum and trend direction
- ADX (Average Directional Index) - measures trend strength

Entry Rule

Buy when:

- MACD crosses above its signal line.
- ADX is greater than 25, indicating a strong trend.

Exit Rule

Sell when:

- MACD crosses below its signal line.

Strategy 2 – Mean Reversion

The Mean Reversion strategy assumes that large deviations from recent average prices will eventually reverse. Together, the RSI and Bollinger Bands help spot when an asset is overextended and potentially due for a price reversal.

The strategy uses:

- RSI (Relative Strength Index) - measures market momentum (the speed and strength of price changes)
- Bollinger Bands - measure market volatility (how much the price is fluctuating from its average)

Entry Rule

Buy when:

- RSI is below 30 (oversold).
- Price closes below the lower Bollinger Band.

Exit Rule

Sell when:

- RSI rises above 70 (overbought).
- Price closes above the upper Bollinger Band.

Strategy 3 – Custom Strategy

When combined to form our custom strategy, the 20-day SMA, RSI, and OBV are indicators from multiple categories that create a comprehensive market analysis toolkit for measuring three core elements: trend direction, price momentum, and buying/selling pressure (volume) and with the goal of reducing false trading signals..

The strategy uses:

- 20-day SMA (Trend) - measures the average closing price of an asset over the last 20 trading days
- RSI (Momentum) - measures market momentum (the speed and strength of price changes)
- OBV (On-Balance Volume) - measures buying and selling pressure by cumulatively tracking an asset's trading volume

This satisfies the assignment requirement of using at least three indicators from at least two different indicator categories.

Entry Rule

Buy when:

- Price is above the 20-day SMA.
- RSI is above 50, indicating positive momentum.
- OBV is increasing, confirming buying pressure.

Exit Rule

Exit when one or more of the entry conditions is no longer satisfied (i.e., price falls below the 20-day SMA, RSI falls below 50, or OBV falls below its 20-day moving average).

Performance Comparison Table

Ticker: AAPL	Run Backtest						
	Return	CAGR	Volatility	Sharpe	Sortino	Max Drawdown	Win Rate
Buy & Hold	103.61%	15.43%	27.62%	0.66	0.95	-33.32%	0.00%
Trend Following	41.46%	7.25%	16.46%	0.51	0.44	-21.26%	45.24%
Mean Reversion	35.68%	6.35%	22.58%	0.38	0.41	-30.10%	75.00%
Custom Strategy	45.29%	7.83%	16.10%	0.55	0.50	-27.95%	54.55%

Figure 1: Backtest Metrics in the UI for AAPL

Note: Buy & Hold performs only one long position and therefore does not have closed trades for a meaningful trade-based win rate.

The backtesting engine supports any ticker with sufficient historical data to compute the required indicators, presenting the results in the UI and terminal.

Discussion of Results

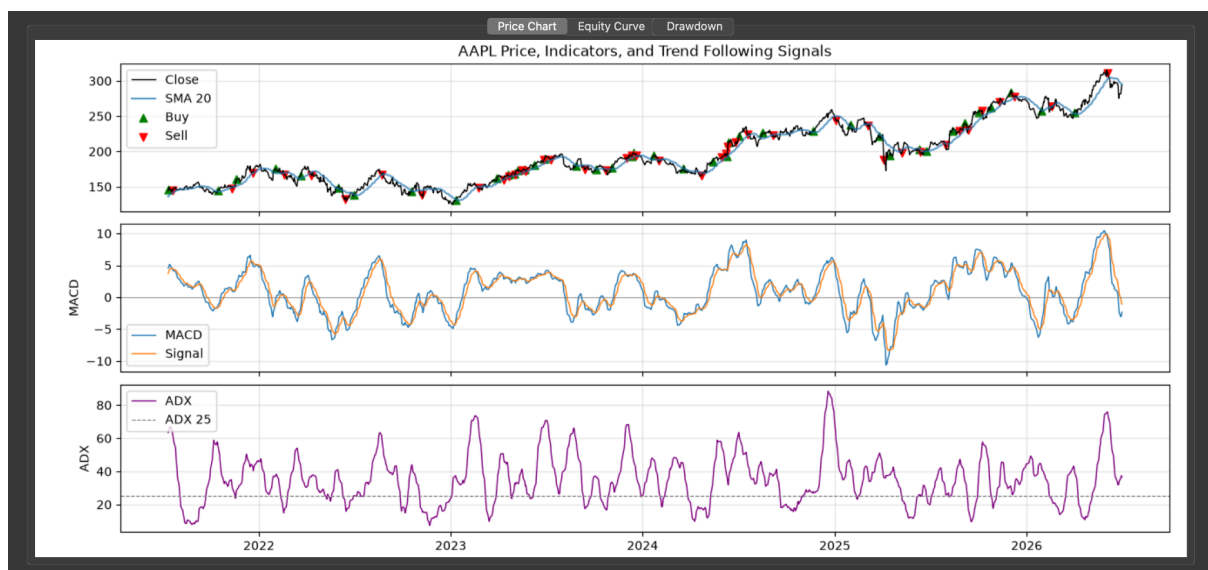


Figure 2: Price Chart for AAPL, including the Trend Following strategy

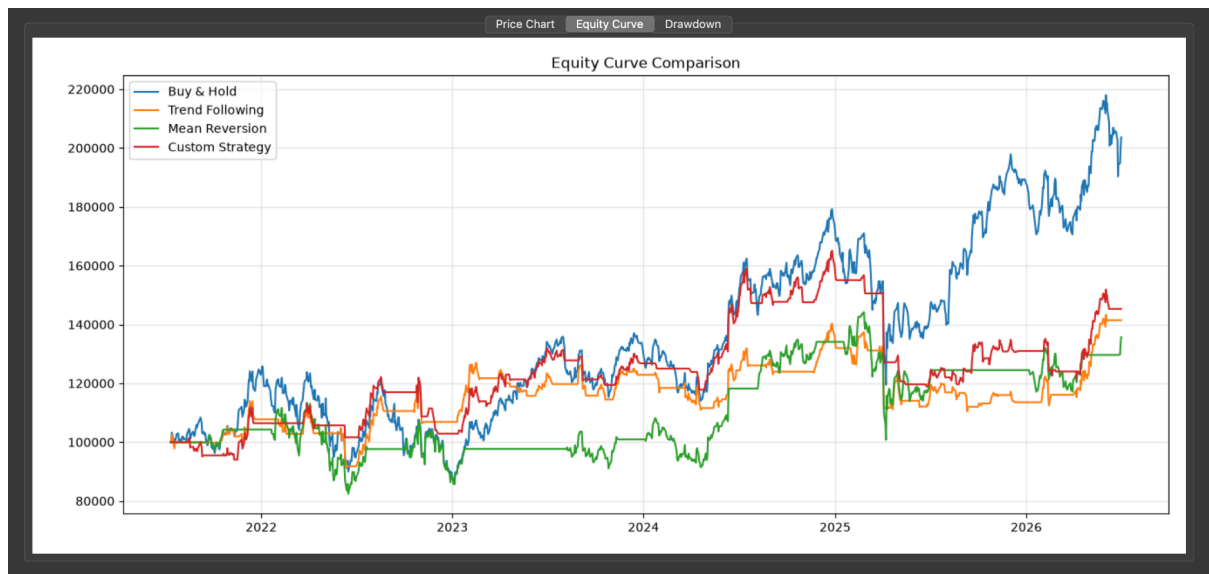


Figure 3: Equity Curve Comparison for all 4 strategies applied to AAPL

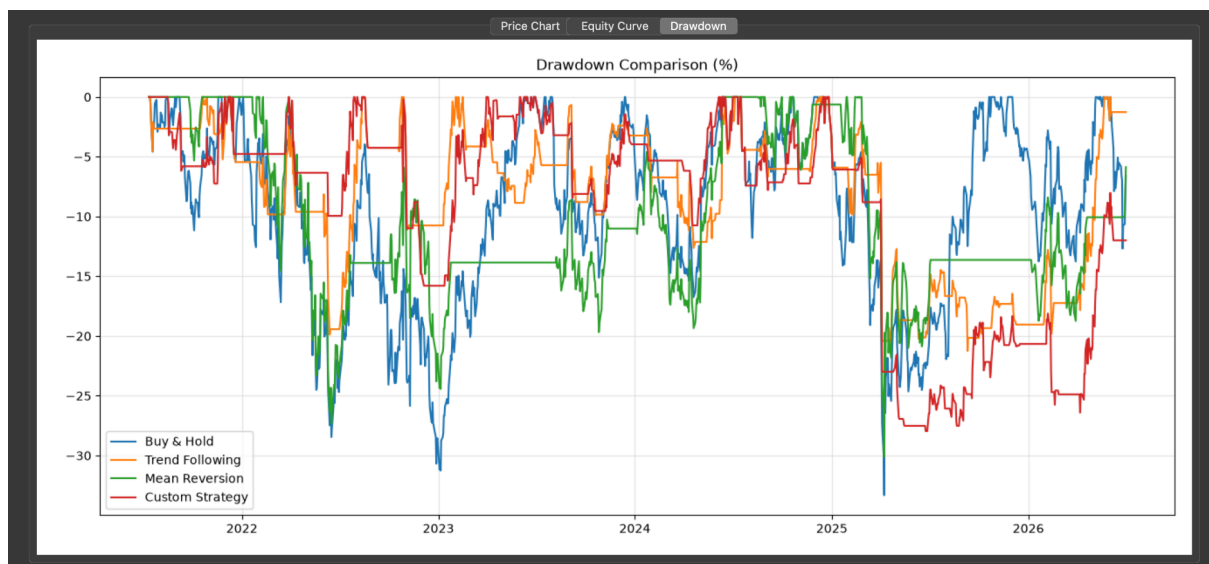


Figure 4: Drawdown Comparison (%) for all 4 strategies applied to AAPL

The Buy & Hold strategy produced the strongest overall performance for AAPL during the five-year backtesting period. It generated a total return of 103.61%, the highest CAGR (15.43%), and the best risk-adjusted performance as measured by both the Sharpe Ratio (0.66) and Sortino Ratio (0.95). These results are consistent with Apple's strong long-term upward trend over the sample period.

The Trend Following strategy significantly reduced downside risk compared to Buy & Hold, achieving the smallest maximum drawdown (-21.26%). However, this came at the cost of substantially lower returns. Because the strategy exits positions whenever momentum weakens, it missed portions of longer bullish trends.

The Mean Reversion strategy produced the highest trade win rate (75%), indicating that most of its trades were profitable. Despite this, its overall return was the lowest among the active strategies. This demonstrates that a high percentage of winning trades does not necessarily translate into the highest portfolio return, since losses can still outweigh many smaller gains.

The Custom Strategy combined trend, momentum, and volume indicators to balance profitability and risk management. It achieved the highest return among the active trading strategies (45.29%) while maintaining better risk-adjusted performance than the Mean Reversion and Trend Following strategies. Although it did not outperform Buy & Hold on AAPL, it demonstrated more balanced behavior by filtering trades using multiple confirmation signals.

Overall, Buy & Hold was the best-performing strategy for AAPL over the selected period, benefiting from the stock's sustained long-term appreciation. Among the active trading approaches, the Custom Strategy offered the strongest combination of return and risk-adjusted performance, while the Trend Following strategy provided the lowest maximum drawdown. The Mean Reversion strategy achieved the highest win rate but the weakest long-term growth, illustrating the trade-off between frequent small gains and overall portfolio performance.